

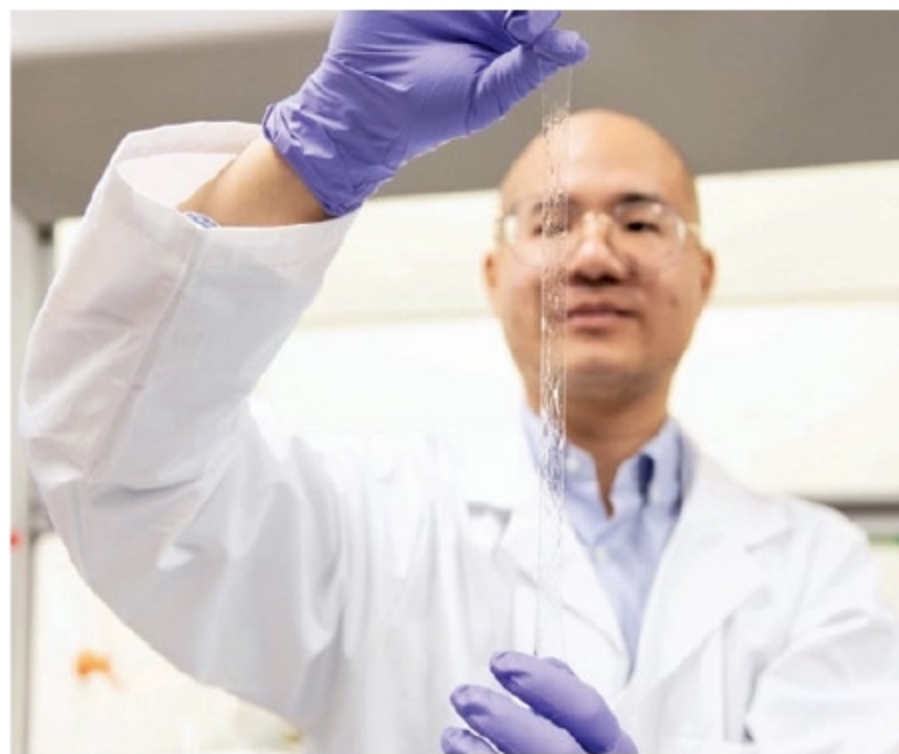
Skin-like sensors bring a human touch to wearable tech

University of Toronto engineering researchers have developed a super-stretchy, transparent and self-powering sensor that records the complex sensations of human skin. Dubbed artificial ionic skin, or AISkin for short, researchers believe the innovative properties of AISkin could lead to future advancements in wearable electronics, personal healthcare and robotics.

Professor Xinyu Liu (MIE), whose lab is focused on the emerging areas of ionic skin and soft robotics, said, "Since it is hydrogel, it is inexpensive and biocompatible—you can put it on the skin without any toxic effects. It is also very adhesive, and does not fall off, so there are so many avenues for this material."

AISkin is made of two oppositely-charged sheets of stretchable substances known as hydrogels. By overlaying negative and positive ions, researchers created what they call a sensing junction on the gel's surface.

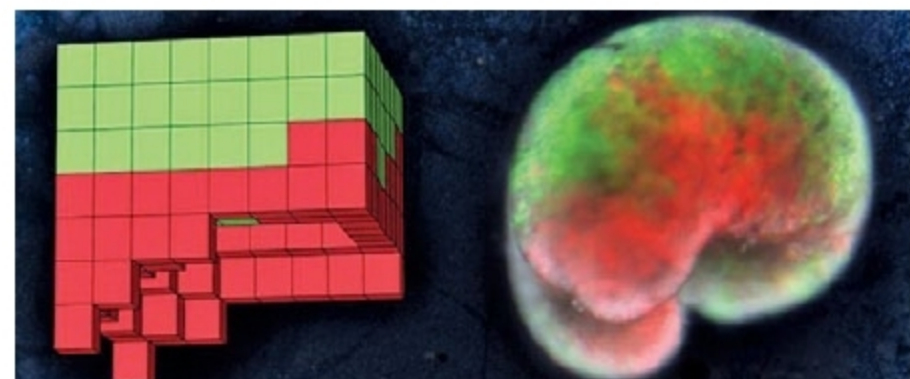
When AISkin is subjected to strain, humidity or changes in temperature, it generates controlled ion movements across the sensing junction, which can be measured as such electrical signals as voltage or current.



AISkin is also uniquely tough and stretchable (Credit: University of Toronto)

Team builds the first living robots

University of Vermont scientists have repurposed living cells—scraped from frog embryos—and assembled them into entirely new life-forms. These millimetre-wide xenobots can move



On the left, the anatomical blueprint for a computer-designed organism, discovered on the supercomputer. On the right, the living organism, built entirely from frog skin (green) and heart muscle (red) cells. The background displays traces carved by a swarm of these new-to-nature organisms as they move through a field of particulate matter (Credit: University of Vermont)

towards a target, perhaps pick up a payload (like a medicine that needs to be carried to a specific place inside a patient) and heal themselves after being cut.

Joshua Bongard, a computer scientist and robotics expert at University of Vermont who co-led the research, said, "These are novel living machines. They are neither a traditional robot nor a known species of animal. It is a new class of artefact—a living, programmable organism."

The new creatures were designed on a supercomputer at University of Vermont, and then assembled and tested by biologists at Tufts University. "We can imagine many useful applications of these living robots that other machines cannot do," said co-leader Michael Levin, who directs Center for Regenerative and Developmental Biology at Tufts, "like searching out nasty compounds or radioactive contamination, gathering microplastic in the oceans, travelling in arteries to scrape out plaque."

Study finds AI can detect changes in patients' mental health

University of Southern California's (USC's) Signal Analysis and Interpretation Lab (SAIL), a centre that focuses on analysing signals by people, from people and for people, in new collaboration with University of California, Los Angeles (UCLA), has found that artificial intelligence (AI) can accurately detect changes in clinical states from speech as well as physicians.

SAIL, which has long applied AI and machine learning to identify and classify video, audio and physiological data, partnered with researchers at UCLA to analyse voice data from patients being treated for serious mental illnesses, including bipolar disorder, schizophrenia and major depressive disorders. These individuals and their treating clinicians used MyCoach-Connect interactive voice and mobile tool, created and hosted on Chorus platform at UCLA, to provide voice diaries related to

their mental health states. SAIL then collaborated with UCLA researchers to apply AI to listen to hundreds of voicemails using custom software to detect changes in patients' clinical states. SAIL AI was able to match clinicians' ratings of their patients.

"Machine learning allowed us to illuminate the various clinically-meaningful dimensions of language use and vocal patterns of patients over time and personalised at each individual level," said senior author Dr Shri Narayanan, Niki and Max Nikias Chair in Engineering and Director of SAIL at USC Viterbi School of Engineering.

Tracking changes in clinical states is important to detect if there is a change that shows that condition has improved or worsened that would warrant the need for changing treatment, according to the researchers.